USB-C: power + native USB (flashing, logs).

3.3 V: SY8089 buck, Vout = 0.6 V x (1 + 100k / 22k) = 3.33 V

J1
USB-C

PWR_FLAG
PWR_FLAG
PWR_FLAG

U2 SY8089AAAC

ESP32-S3-WROOM-1-N16R8

4.3" 800x480 RGB panel, standard 40-pin ribbon.
RGB on GPIOs in connector order (B-G-R, 5-6-5 bits, low bits to GND).

Scanner on UART0: SparkFun 2D Barcode Scanner (3.3 V)
or a 5 V UART module on J5. Fit only one.

SparkFun 2D Barcode Scanner

Backlight: TPS61169 boost, I = 204 mV / R13 = 20 mA.
PWM dimming from IO5. 37.5 V overvoltage limit.

U5 TPS61169

UART scanner (GM65)

GT911 touch ribbon (6-pin, 1.0 mm)

4.3" RGB panel

GT911 touch

Buzzer: MLT-8530 from 5 V, PWM on IO2.

Status LED: side-view SK6812 on the bottom edge, data 3.3 V -> 5 V.

Generated by tools/make_sch.py from tools/design.py

Sheet: /
File: mainboard.kicad_sch

Title: Pantry wall scanner mainboard

Size: A3	Date: 2026-10-27	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 1/1